

BONSEO KOO

Undergraduate Researcher in Electronic Engineering

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Research Interests: AI Accelerators · Digital VLSI · Low-Power Digital Systems · FPGA/SoC · HW/SW Co-Design

EDUCATION

Sogang University — B.S. Candidate in Electronic Engineering

Mar. 2021 – Present

Six semesters completed | GPA: **3.79/4.50** | Official percentile: **92.4%**

- Selected coursework: Digital Circuit Design (A+), Digital Logic Circuit Laboratory (A0), Computer Architecture (A0), Advanced Digital Circuit Design (A0), Microprocessor, Probability and Random Variables (A0).

RESEARCH EXPERIENCE

Undergraduate Research Intern — Yonsei University, Department of Electrical & Electronic Engineering

Summer 2026

Advisor: Prof. Seungkyu Choi

- Completed a LeNet-based MNIST inference NPU design project on a Zynq SoC, extending experience from PL-only accelerator design toward PS–PL system integration.
- Developed PS-side control in Vitis and worked with AXI4-Lite-based memory-mapped control/data access and DMA-based data transfer for FPGA accelerator integration.
- Completed a research-paper presentation on “OliVe: Accelerating Large Language Models via Hardware-friendly Outlier-Victim Pair Quantization,” analyzing low-bit quantization, outlier handling, accelerator performance, and energy efficiency.

FPGA NPU Architecture Design & Optimization — Sogang University

Dec. 2025 – Jul. 2026

Advisor: Prof. Sungju Ryu

- Designed and implemented an FPGA NPU in Verilog RTL based on an 8×8 weight-stationary systolic array and matrix-tiling controller; validated the design on PYNQ-Z2.
- Proposed LOD-based Dynamic Scaling and Switching-Aware Zero-Skipping to improve the fidelity–area–power trade-off of low-precision inference.
- Evaluated 45 NPU configurations and selected M3_no8 through fidelity, hardware-cost, and dynamic-power analysis.
- Integrated the FPGA NPU into a live-camera CIFAR-10 inference pipeline and demonstrated the system at the 2026 ISE Summer Conference.

PUBLICATION & CONFERENCE PRESENTATION

B. Koo, Y. Lee, T. Jung, and S. Ryu, “Maximizing NPU Fidelity and Efficiency with LOD-Based Dynamic Scaling and Switching-Aware Zero-Skipping,” *2026 ISE Summer Conference*, first-author oral presentation, Jul. 2026.

- Presented the work in an undergraduate oral session and a separate live FPGA demonstration session.

AWARDS

Live Demonstration Excellence Award — 2026 ISE Summer Conference

Jul. 15, 2026

- Awarded for the live FPGA demonstration of the low-precision NPU and CIFAR-10 inference pipeline.

SELECTED ENGINEERING PROJECTS

LeNet-Based MNIST Inference NPU — Yonsei University

2026

- Implemented a CNN inference accelerator project with Zynq PS–PL integration, Vitis-based PS software, AXI4-Lite control, and DMA-based data movement.

Digital Logic Circuit Laboratory — Wristwatch-Style Digital System

2025

- Designed a wristwatch-like digital system using a 7-segment display and an 8×8 LED matrix, with emphasis on logic optimization and hardware implementation.

TECHNICAL SKILLS

Hardware / RTL: Verilog HDL, RTL Design, FPGA Design, Systolic Arrays, NPU Architecture, RTL Simulation & Verification

FPGA / SoC: AMD/Xilinx Vivado, Vitis, PYNQ-Z2, Zynq PS–PL Integration, AXI4-Lite, DMA, BRAM

Programming / ML: Python, PyTorch, Jupyter Notebook, C

Other Tools: MATLAB, Cadence Virtuoso, PSpice, TCAD, Visual Studio

LANGUAGE

TOEFL iBT: 5.0/6.0 overall (100/120 equivalent; CEFR C1) — Reading 5.0, Listening 5.0, Writing 4.0, Speaking 5.0